Neonatal Sepsis

Saturday, January 8, 2022 2:26 PM

❖ Definition
- Clinical syndrome of bacteremia with systemic sign and symptom of infection

❖ Classification

➢ **Early sepsis**
- Acquired before or during delivery (vertical mother to child transmission)
- In the first week usually(less than 72 hr)

○ **Risk factor**
- Prematurity
- Premature rupture of membrane more than 18 hr
- Chorioamnionitis
- Maternal intepartum fever

○ **Common organism:**
- Group B streptococci
- E.Coli
- Listeria monocytogenes

➢ **Late sepsis**
- Develop after delivery from organism acquired in the hospital or community
- After the first week until 3 month of age

○ **Risk factor:**
- Prematurity
- Hospitalization
- Umbilical catheterization
- Endotracheal intubation
- Mechanical ventilation

○ **Common organism:**
- Staphylococcus Aureus
- Hemophilus influenza
- Kelbsiella
- Pseudomonas
- Viral or candida

❖ Clinical picture

➢ **Early manifestation**
- Lethargy (excessive sleepiness)
- Poor feeding or feeding intolerance
- Unstable temperature (hypothermia more common and serious)
- Apneic attack ,pallor, cyanosis
- Poor Moro and suckling reflex

➢ **Late manifestation**
- **Early manifestation plus more focal infection**
- Meningitis
- Pneumonia
- Direct hyperbilirubinemia
- Necrotizing enter colitis
- Septic shock +
- Septic renal failure with oliguria and metabolic acidosis
- Purpura and DIC
- sclerema: Harding of the skin (Poor prognostic sing)

❖ Diagnosis
- History
- Clinical picture
- investigation

○ **investigation**
- **Sepsis screening**
  - CBC finding: leucopenia less than 5000/mm (with sever sepsis)
  - Pandemia (pan cell immature) more than 20% of total neutrophil count
  - Less common leukocytosis more than 30000/mm
  - Thrombocytopenia
- **Inflammatory marker**
  - CRP
  - ESR
- **culture**
  - Detection of causative organism by culture of blood, CSF ,urine, Swab culture (orifice)
- **Evidence of multiorgan systemic disease**
  - Respiratory ↦ x ray for pneumonia
  - CNS ↦ CSF analysis for meningitis
  - Liver ↦ enzyme , bilirubin and PTT
  - Kidney ↦ Serum urea and electrolyte

❖ Differential diagnosis ↦ THE MIS FITS
- Trauma e.g. (Intracranial hemorrhage)
- Heart disease e.g. (Congenital ,hypoxia , hypovolemia)
- Endocrine e.g. (Congenital adrenal hyperplasia)
- Metabolic disturbance e.g. (hypoglycemia , hypocalcemia)
- Inborn errors of metabolism
- Sepsis
- Fits (Seizers)
- Intestinal catastrophes e.g. intestinal obstruction
- Toxins
- Sever asphyxia

❖ Management

➢ Incubator care in neonatal care unit(**NICU**)

➢ **Antibiotic** are given according to culture and sensitivity

- While waiting for C/S: empirical Antibiotic given
- Penicillin(Ampicillin or cloxacillin) plus Aminoglycoside (gentamycin)
- third generation cephalosporin such as cefotaxime ceftazidime
- All antibiotic should given for 10_14 days(3weeks for meningitis)

❖ Prevention

- Exclusive breastfeeding
- Keep cord dry
- Hygiene of baby
- No unnecessary interventions
- **Control of hospital infections**
  - Hand washing by all staff
  - Isolation of infectious patient
  - Avoid overcrowding